The border brief: an ecosystem evidence web

Teacher guide and worked answers

Describe an ecosystem and biodiversity, then use a food web and sample counts to support a cautious recommendation.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C36

Describe ecosystems & biodiversity.

Directly develops the stated outcome. Original teacher-authored exemplar; route support is responsive and not a qualification or fixed ability label.

Prior learning

Explain that plants make organic substances through photosynthesis.

Read a simple data table and recognise that samples can vary.

Success evidence

Distinguish a community from an ecosystem by including abiotic interactions.

Read food-web arrows from food to consumer and predict a possible indirect effect.

Compare species richness fairly without confusing it with total abundance or proof of cause.

Equipment

Shared evidence sheet, one selected response route and exit sheet; pencils.

Preparation

Read the worked answers and preview the three interactive decisions.

Print shared page 1, one route from pages 2-4, and page 5.

Explain that the food web is a simplified garden model and the counts come from a separate plant-sampling task. They are not measurements of the animals in the web.

Practical care

The complete route uses printed evidence at desks. No live animals, plants or soil need handling.

Optional follow-up sampling stays within an agreed accessible area; pupils do not taste plants, handle unknown organisms or disturb nests. Wash hands after outdoor work.

Ready to teach alternative

Use the supplied fictional garden web and matched-area constructed count data. Pupils can make a complete evidence brief within 40 minutes without collecting outdoor samples.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Which border deserves a closer look?
2	Retrieval	2-5	Build the starting vocabulary
3	I do	5-13	I explain an interaction
4	I do	Within stage	I read the arrow in the right direction
5	I do	Within stage	I count species separately from individuals
6	We do	13-21	The garden evidence desk
7	We do	Within stage	Predict a possible knock-on effect
8	We do	Within stage	Write a claim the evidence can carry
9	Hinge	21-24	Choose the defensible headline
10	You do	24-34	Write the garden team’s brief
11	You do	Within stage	Check your arrows
12	You do	Within stage	Check your evidence language
13	Review	34-38	Peer-review the brief
14	Review	Within stage	Give the garden team a question
15	Exit	38-40	Make three ideas precise

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Which border deserves a closer look?

Introduce a fictional garden decision. More individuals and more species are different claims. Ask pupils to predict what else they need to know before supporting a decision.

2 Retrieval Build the starting vocabulary

R1 photosynthesis. R2 non-living. R3 no; a sample represents part of a site and may miss species. Revisit sample versus population if needed.

3 I do I explain an interaction

Think aloud: light is abiotic. Its availability can affect plant photosynthesis. Plant cover can change shade and moisture conditions. An ecosystem explanation includes relationships, not just two disconnected lists. Soil contains living and non-living components; soil moisture is the non-living factor here.

4 I do I read the arrow in the right direction

Shared I do time. Trace leaves → aphids → ladybirds and leaves → caterpillars → blue tits. The model selects plausible garden feeding links and omits decomposers and other foods. It does not predict exact population sizes or timings.

5 I do I count species separately from individuals

Shared I do time. Work the analogous mini-example aloud: sample X contains 3 daisies and 3 clover plants, so 6 individuals and 2 species. For the displayed main data ask pupils to do their own counting during We do. A zero means not recorded in this sample, not proven absent everywhere.

6 We do The garden evidence desk

8 minutes across three We do steps. First classify soil moisture as abiotic and aphid feeding as biotic, then connect one factor to an organism. All necessary web links and counts are printed on shared page 1.

7 We do Predict a possible knock-on effect

W2: ladybirds may have less prey and their abundance may fall, but alternative foods, movement, timing and other factors could alter outcomes. Blue tits also have other foods in the simplified web. Do not use numerical reductions or inevitable extinction.

8 We do Write a claim the evidence can carry

W3 A=2 recorded species, B=4, so B greater recorded richness in these samples. Totals equal. Cannot establish site-wide biodiversity or cause from one sample per patch. Teacher helps replace “proves” with “in these samples”.

Reveal: Patch B: 4 recorded species. Patch A: 2. One sample cannot establish the cause.

9 Hinge Choose the defensible headline

A incorrect: abundance and richness differ. B correct, bounded by the samples. C incorrect: one sample and no controlled comparison cannot prove cause. If uncertain, count non-zero species separately from summing individuals.

A. Both patches have identical biodiversity because totals match. - Equal abundance does not imply equal richness or all other biodiversity measures.

B. Patch B has greater recorded species richness in these samples. - Four species were recorded in B compared with two in A.

C. The mowing method definitely caused the difference. - The data do not isolate mowing or rule out sampling variation.

10 You do Write the garden team’s brief

10 minutes independent. Core evidence is an ecosystem relationship, correctly directed feeding link, and bounded richness comparison. Pupils may draw with labels or dictate their own analysis. A recommendation is to investigate further, not to claim a proved cause.

11 You do Check your arrows

Shared You do time. Prompt checking without correcting individual arrows before they attempt the task.

12 You do Check your evidence language

Shared You do time. Repeated matched-area plant samples at randomly selected locations are an appropriate next step. Measuring another relevant factor can help interpret differences.

13 Review Peer-review the brief

Offer teacher or private self-review as alternatives to peer review. Accept distinct accurate abiotic links. Do not penalise pupils for choosing a cautious recommendation.

14 Review Give the garden team a question

Lundy: capture pupils’ questions, choose one feasible follow-up and explicitly report how their reasoning influenced the next task. Do not require pupils to promise outdoor participation.

15 Exit Make three ideas precise

E1 interactions with abiotic environment. E2 transfer from food to consumer. E3 no; need counts split by species. E4 pupil sheet asks one sampling improvement. Collect before model answer.

Answers and assessment

R1-R3

R1 photosynthesis. R2 soil moisture is non-living/abiotic. R3 one sample is incomplete and may miss organisms or species.

W1-W3

W1 e.g. soil moisture affects plant growth, or light affects photosynthesis; aphid feeding is biotic. W2 fewer aphids may reduce food for ladybirds and lead to fewer ladybirds, but alternative prey, movement or other factors can affect the outcome. W3 both have 20 individuals; A records 2 species and B 4. B has greater recorded richness in these samples. These data do not prove a cause or full-site biodiversity.

Hinge A-C

A wrong: equal abundance does not mean equal richness. B correct and limited to the evidence. C wrong: sampling variation and confounding factors are unresolved.

S1-S5

S1 soil moisture. S2 non-living. S3 correct arrow sequence; leaves are eaten by aphids, aphids by ladybirds, transferring material and energy. S4 two/four/twenty. S5 samples vary and one small area may miss species; more comparable samples improve the estimate.

N1-N4

N1 abiotic: light→photosynthesis or soil moisture→plant growth; biotic: feeding or a justified competition example. N2 e.g. leaves → aphids → ladybirds and leaves → caterpillars → blue tits. Fewer aphids may reduce ladybird food; other prey/movement/interactions create uncertainty. N3 abundance 20 both; richness A 2 / B 4. N4 repeat equal-area samples at randomly selected locations in both patches using comparable identification and effort; one unmatched observational comparison does not isolate maintenance from other factors.

T1-T3

T1 the count measures recorded plant richness in small samples, not all biodiversity in the whole garden; causal attribution to mowing is unsupported. Accurate rewrite: “Our equal-area sample from B recorded four plant species, compared with two in A; further sampling is needed.” T2 sample several randomly selected equal-area quadrats in each defined patch, at comparable times with the same counting/identification rules and equal effort; record relevant conditions. To test mowing causally, use replicated comparable plots with assigned regimes and before/after measurements where feasible. T3 aphid feeding damage may fall; ladybirds may lose prey; blue tits could lose aphids as prey but have other food. The web omits many foods, movements, abundance and timings, so exact size or certainty cannot be inferred.

E1-E4 / reflection

E1 abiotic environment and interactions with living organisms. E2 ladybird consumes aphid; transfer from food to consumer. E3 individuals can be distributed among different numbers of species. E4 repeated randomly located equal-area samples reduce selection bias and show variation; common method/equal effort improve comparability. Reflection accepts a relevant evidence-seeking question and reason.

Evidence and responsive teaching

Secure core: explains an ecosystem interaction, reads food-to-consumer arrows, and gives A 2 / B 4 richness while recognising abundance 20 each and a sampling limit. Re-teach reversed arrows with “is eaten by”; re-teach richness confusion by counting only rows with non-zero values before adding individual counts. Formal diversity indices, trophic efficiency and population equations are not required.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Print shared page 1, ONE selected response route (2, 3 or 4), and exit page 5. Do not require all routes. Adult scribing records the pupil’s own words.

This is an introductory Topic 9 lesson after the planned Topics 7-8 assessment. Later lessons can develop quadrats, transects, abundance estimates and biodiversity measures.

The selected food web and the separate plant dataset serve different reasoning tasks. Do not imply the plant counts measured the animals in the web.

No real school biodiversity claim is made. The constructed data intentionally keep total abundance equal to isolate the richness distinction. Zero means not recorded in that sample, not certain absence at the site.

Teacher vocabulary extension: biodiversity includes genetic, species and ecosystem variation; species richness is one useful but incomplete measure. Greater richness alone does not establish a healthier site in every ecological context.

Access and responsive teaching

Supported

Supply short vocabulary choices, an arrow sentence and counts to circle. Accept a labelled evidence map plus spoken reason.

Standard

Create a garden brief linking environment, feeding relationships and sample evidence.

Stretch

Evaluate sampling limits and propose matched repeated measurements to test a maintenance decision. No diversity-index calculation required.

Regulation

Use a fictional garden decision. Allow quiet written votes and private review. Choose route from retrieval and hinge responses; pupils can change route when appropriate.

Misconceptions to check

An ecosystem is just a list of animals.

It includes the community, non-living environment and their interactions.

Check: Ask where light and soil moisture fit.

Food-web arrows show the predator pointing at its prey.

Arrows show transfer of material and energy from food to consumer.

Check: Read leaf→aphid aloud.

Twenty organisms always means greater biodiversity than ten.

Abundance alone does not tell us how many species are present.

Check: Compare two samples that both contain twenty plants.

One sample proves that a mowing regime caused the difference.

Sampling can vary; other conditions may differ. Repeat comparable sampling and use a fair design before causal claims.

Check: Ask what the sample comparison cannot establish.

Word	Meaning
population	Organisms of one species in a particular area.
community	Populations of different species living in an area.
ecosystem	A community interacting with the non-living environment.
biotic	Living components and effects of organisms, such as predation.
abiotic	Non-living environmental factors, such as light or moisture.
biodiversity	The variety of life; here we compare species richness as one aspect.
species richness	The number of different species present in a stated sample.
abundance	The number of individuals present in a stated sample.

Scientific references

OpenStax Biology 2e: Ecology of ecosystems

Checked September 2026. Ecosystem interactions and food-to-consumer arrows; our selected garden links and tasks are original.

OpenStax Biology 2e: Community ecology

Primary-source reference checked by the independent audit. Species richness and relative abundance are distinct community measures.

OpenStax Biology 2e: The biodiversity crisis

Checked September 2026. Biodiversity includes variation at more than one level; richness is only one aspect.

Pearson Edexcel GCSE Biology 1BI0 specification

Actual Issue 4 PDF checked. Topic 9 ecological relationships and sampling progression; this introduction is an original teacher resource, not an official assessment.